(see 6.7)

MILITARY SPECIFICATION

DETONATOR ELECTRIC M95 (T61) PARTS FOR LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing for one type of electric detonator designated as the M95 Electric Detonator.

2. APPLICABLE DOCUMENTS

21. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-559 - Ink, Marking, Stencil, Opaque, for Porous Surfaces, Wood Boxes, Fiber Cartons, etc.

MILITARY

MIL-P-116 - Packaging and Packing for Overseas Shipment
MIL-B-117 - Bags, Interior Packaging
MIL-A-148 - Aluminum Foil
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1375

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MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8816368 - Disc, Lead
8871907 - Holder, Assembly
8873302 - Operating Instructions
9200356 - Set-up functioning
9200602 - Bushing
9200657 - Disc centering
9205891 - Detonator, Electric, M95 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, 25, D.C. 20402. Orders shall specify 49 CFR 171-179 (latest revision).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonators. - The detonators shall comply with all requirements specified on Drawing (dwg. 9205891, and with all requirements specified in applicable specifications.

3.3 Moisture content.

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3.3.1 Lead azide. - The moisture content of the lead azide at the loading station at the time of loading the detonator shall not exceed 0.50 percent.

3.3.2 PETN. - The moisture content of the PETN at the loading station at the time of loading the detonators shall not exceed 0.50 percent.

3.4 Resistance between shorted lead wires and cup. - The resistance between the shorted lead wires and the cup shall be not less than 50 megohms.

3.5 Waterproofness. - The detonator shall be waterproof as evidenced by normal functioning after being immersed in water maintained at a temperature of 70 degrees Fahrenheit (F.) plus or minus 10 degrees F. for a period of 48 hours, minimum, at a depth of 2 to 3 inches.

3.6 Functioning. - The detonators shall comply with the following:

3.6.1 Time. - The detonators shall fire on the first attempt within 50 microseconds.

3.6.2 Output. - The detonators shall fire on the first attempt and shall make a 3/16 inch minimum (min.) perforation in a .1345 inch thick lead disc.

3.6.3 Duds. - The detonators shall not fail to fire on the first attempt.

3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.8 First article inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings for the moisture content of each explosive shall be submitted with each lot submitted for acceptance.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the cerfitying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation as specified herein. The first article sample shall consist of inert parts (see 4.2.2) and loaded detonators (see 4.2.3) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inert Parts. - A sample of 25 each of every part shall be submitted in accordance with the provisions of 4.2.1 for inspection as specified in 4.2.2.1.

4.2.2.1 Inspections to be Performed. - The parts will be subjected by the Government to any or all of the applicable examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings.

4.2.2.2 Rejection. - If any part or assembly fails to comply with any of the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon the failure of an item to comply with any of the requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.2.3 Loaded Detonators. - A sample of 185 loaded detonators shall be submitted in accordance with the provisions of 4.2.1 for testing as specified in 4.2.3.1.

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4.2.3.1 Test to be Performed. - The loaded detonators shall be tested as specified in 4.3.3.6, Table II.

4.2.3.2 Rejection. - The loaded detonator first article sample shall be rejected if the sample fails to meet the criteria of 4.3.3.6.2. In the event of rejection, the Government reserves the right to terminate its inspection upon failure of any detonators to comply with the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied, to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonators shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.
b. PETN of one lot from one supplier.
c. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision, and which have subsequently been subjected to uniform controlled drying and operating conditions at the loading plant.

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4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL Of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 9206001 covering a detail of dwg. 9205891)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Thickness through bottom	Gage	01001
102.	Cup dented or distorted	Visual	01002
Minor:	AQL 1.00 percent		
201.	Outside diameter, min.	Gage	01003
202.	Evidence of poor workmanship (see 3.7)	Visual	01004

4.3.2.2 Plug and wire assembly (see dwg. 9205894 covering a detail of dwg. 9205891).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Large outside diameter min.....	Gage	02001
102.	Length of lead wires, min.	Gage	02002

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103.	Length of insulation on lead wire, max.	Gage	02003
104.	Distance beween wires, lead end, min.	Gage	02004
105.	Plug cracked or chipped	Visual	02005
106.	Lead wires not tinned	Visual	02006
Minor:	AQL 1.50 percent		
201.	Length of plug	Gage	02007
202.	Height of plug boss	Gage	02008
203.	Evidence of poor workmanship (see 3.7)	Visual	02009

4.3.2.3 Bridge and plug assembly (see dwg. 9205895 covering a detail of dwg. 9205891).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Crack, void or separation in spot charge	Visual	03001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03002

4.3.2.4 Detonator, electric, M95 prior to inserting bridge and plug assembly (see dwg. 9205891).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of base charge	Gage	04001
Minor:	aQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	04002

4.3.2.5 Assembly (see dwg. 9205891).

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Lead wires not shunted	Visual	05001
2.	Insulation not removed on shunted lead wires	Visual	05002
3.	Cup cracked or split	Visual	05003
Major:	AQL 0.65 percent		
101.	Length	Gage	05004
102.	Diameter	Gage	05005
103.	Crimp not full 360 degrees or not flat against plug	Visual	05006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05007

4.3.2.6 Carton, prior to sealing (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Incorrect number of detonators..	Visual	06001
102.	Detonators in bundle not completely encased in foil	Visual	06002
103.	Packet assembly torn or insecurely closed (when applicable)	Visual	06003
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	06004
202.	Package marking missing, incorrect or illegible	Visual	06005

4.3.2.7 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Carton damaged to extent that contents are exposed or liable to become exposed	Visual	07001
Minor:	AQL 0.65 Percent		
201.	Marking missing, incorrect or illegible	Visual	07002

4.3.2.8 Sealed Carton in Sealed Barrier Bag (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Bag punctured or separation of heat seal	Visual	08001
Minor:	AQL 0.65 Percent		
201.	Marking missing, incorrect or illegible	Visual	08002

4.3.3 Production Testing. - Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with Standard MIL-STD-105.

4.3.3.1 Moisture Content (see 3.3) - Major Defect

TABLE I

Explosives	Code No.
Lead Azide	09001
PETN	09002

The contractor shall provide adequate controls to insure that the explosives comply with the requirement. For verification, the contractor shall select, and subject to test, a sample from that quantity of each explosive used in each eight (8) hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has

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not begun, the explosive represented by the sample shall be rejected. If components have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1 and 4.4.2.

4.3.3.2 Pull Test of Lead Wires (see dwg. 9205894 covering a detail of dwg. 9205891) - Major Defect - Code No. 02010.

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Code Letter J and an AQL of 0.65 percent. Any plug assembly that fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.3.

4.3.3.2.2 After three consecutive lots have complied with the criteria of 4.3.3.2.1, the sampling plan shall be in accordance with Code Letter G and an AQL of 0.40 percent. Any plug assembly that fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.3.

4.3.3.3 Resistance Between Lead Wires (see dwg. 9205895 covering a detail of dwg. 9205891) - Major Defect - Code No. 10001. - This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.4 Resistance of Lead Circuit (see dwg. 9205891) - Major Defect - Code No. 10002. - This test shall be performed 100 percent in accordance with the procedure specified in 4.4.5. Any assembly that fails to comply with the requirement shall be classed defective ad removed from the lot.

4.3.3.5 Resistance Between Shorted Lead Wires and Cup (see 3.4) - Major Defect - Code No. 10003. - This test shall be performed 100 percent in accordance with the procedure specified in 4.4.6. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.6 Functioning and Waterproofness

4.3.3.6.1 Sampling. - One hundred eighty-five (185) detonator assemblies shall be selected from each lot for test. The

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sequence of testing and number of samples to be subjected to each test is specified in Table II.

TABLE II

Sequence - Test	Classification	Code No.	No. of Samples	Test Paragraph
1. Time (see 3.6.1)	Major	05008	35	4.4.7.1
2. Waterproofness & Output (see 3.6.2, 3.5)	Major	05009	50	4.4.7.2
3. Duds (see 3.6.3)	Major	05010	*100 less time and output duds	4.4.7.3

* Duds that occur in the time or output phase of testing shall be recorded and replaced from this sample.

4.3.3.6.2 Rejection. - The lot shall be rejected if any of the following test results occur:

a. Time. - The mean functioning time ($\overline{X}$) plus the product of 2.27 and the standard deviation (s) (see 6.5) exceeds 50 microseconds.

b. Waterproofness and Output. - One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.

c. Duds. - The total number of detonator assemblies that fail to fire on the first attempt (dud) in the time, output and dud tests equals or exceeds two (2).

4.3.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" in 4.3.2 and 4.4. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

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4.4 Test Methods and Procedures

4.4.1 Moisture Content of Lead Azide. - A sample of approximately two (2) grams (gm.) of lead azide accurately weighed to the nearest .0001 gm. shall be placed in an uncovered tared Pyrex dish approximately 55 mm in diameter and 30 mm in height. The weighed sample and dish shall be dried for 1 1/2 hours in an oven maintained at 95 plus or minus 2 degrees Centigrade (C.). At the end of the drying period, the sample shall be removed from the oven and immediately placed in a desiccator containing anhydrous calcium chloride to cool for approximately 20 minutes. After cooling, weigh the dish and contents to the nearest .0001 gm. and calculate the percentage of moisture in the sample from the loss in weight.

4.4.2 Moisture Content of PETN. - The moisture content of the PETN shall be determined as specified in 4.4.1.

4.4.3 Pull Test of Lead Wires. - This test shall be performed using the equipment specified in 4.3.4. After application of the required force to each lead wire, observation shall be made for slippage at the entry point of the wires into the plug and on the opposite face of the plug. Any unit exhibiting any slippage whatsoever shall be classed defective.

4.4.4 Resistance Between Lead Wires. - The plug assembly shall be connected to the applicable test fixture specified in 4.3.4 and the voltage as specified on the applicable drawing shall be applied. If the resistance between the lead wires is less than that specified on the applicable drawing, the plug assembly shall be classed defective.

4.4.5 Resistance of Lead Circuit. - Prior to shorting the lead wires, the resistance of the detonator shall be measured with an electrical ohmmeter using not more than 20 microampere of current. Any assembly which fails to comply with the applicable requirement shall be classed defective.

4.4.6 Resistance Between Shorted Lead Wires and Cup. - The detonator shall be connected to the applicable test fixture specified in 4.3.4 and a minimum potential of 500 volts 60 cycles (R.M.S.) shall be applied for two (2) seconds maximum between the shorted lead wires and the cup. Any assembly which fails to comply with the specified requirement shall be classed defective.

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4.4.7 Functioning. - Detonator assemblies used for the following tests shall be temperature conditioned to 70 plus or minus 10 degrees Fahrenheit (F.) for two (2) hours minimum prior to testing. Method of inspection shall be in accordance with dwgs. 8816368, 8871907, 8873302, 9200356, 9200602 and 9200657.

4.4.7.1 Time. - The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy from a .004 microfarad capacitor charged to 100 volts maximum. The elapsed time between application of the energy and explosion shall be measured.

4.4.7.2 Waterproofness and Output. - The detonator shall be submerged in water for the time specified, and at the depth and temperature specified. After waterproofness, the detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.7.1. The diameter of the perforation made in the lead disc shall be measured.

4.4.7.3 Duds. - The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.7.1. Observation shall be made to determine compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level B

5.1.1.1 Rosette Wrap. - Twenty-five (25) detonators with twisted lead wires out straight shall be made into a bundle using the lead wires of one detonator to secure the bundle. Each bundle shall be wrapped in approximately an eight (8) by eight (8) by .0015 inch thick size sheet of aluminum foil complying with Specification MIL-A-148, so that all detonators and lead wires in a bundle are completely encased in the foil. Five (5) bundles of detonators (125 detonators) shall be packed in a folded carton set-up box, or comparable container, made of material with a minimum board thickness of 0.03 inch. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of assembly into end items, the following shall be included: Each carton, set-up box or comparable container, after blunting corners, shall be placed

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in a water and vapor proof gab manufactured in accordance with Type I, Class E of Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of Specification MIL-P-116.

5.1.1.2 Pocket Assembly, Alternate Pack. - The detonators shall be packed in pocket assemblies manufactured in accordance with Figure 1 (as applicable) made from .0015 inch thick aluminum foil complying with Specification MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five (5) pockets. The strength of the seal shall be sufficient to allow insertion of the dowel (see Figure 1) without separation of the seal. The size of each pocket between seals shall not vary by more than 1/16 inch. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end (see Figure 1). Any alternate method that results in a satisfactory pocket without damage to the material may be used. One (1) M95 Electric Detonator Assembly shall be inserted, detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to close the pouch. Twenty-five (25) closed pocket assemblies containing a total of 125 detonators shall be packed in a folded carton, set-up box, or comparable container made of material with a minimum board thickness of 0.03 inch. If shipment and uncovered storage is to exceed 180 days, the detonator container will be protected as specified in 5.1.1.1.

5.1.2 Level C. - Unless otherwise specified, the M95 Electric Detonators shall be preserved and packaged as specified in 5.1.1 except that the barrier bag is not required.

5.2 Packing

5.2.1 Level B. - A suitable number of containers, each containing 125 detonator assemblies, shall be packed in an outer case in accordance with the Code of Federal Regulations CFR 49 171-179 requirements for electric blasting caps and in a manner that will insure safe delivery.

5.2.2 Level C. - Unless otherwise specified, M95 Electric Detonators shall be packed as specified in 5.2.1.

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5.3 Marking. - Each carton shall be printed, stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. 125 Detonator, Electric, M95

Lot No. Date Packed (Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to Specification TT-I-559.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with dwg. 8796522. The ICC designation shall be (see NOTE):

Electric Blasting Caps - Handle Carefully

e. NOTE: "The number of caps packed in one outer pack shall be shown in the marking."

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection"

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headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval

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to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of Test Data and Ammunition Data Cards. - In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all functioning test data and ammunition data cards shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SMUPA-QA-A-M
Dover, New Jersey 07801

6.5 Calculation of Standard Deviation. - The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma (X - \overline{X})^2}{n - 1}}$$

where:

s = Sample standard deviation
X = Numerical value of individual observations
$\overline{X}$ = Average value of the sample observations
n = Sample size

6.6 Check Test for Possible Deterioration. - The following test is designed to prevent major ammunition items being loaded with detonators that may have deteriorated. This test shall be performed on detonators from each lot which have been stored under normal conditions for more than two (2) years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require a check test, the Contracting Officer shall insert into the contract a check test performance provision. The Contracting Officer will arrange for the contractor

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to be reimbursed for expenses incurred in the performance of the deterioration check test. Two hundred (200) detonator assemblies shall be selected and tested as follows:

6.6.1 Function Time and Duds

6.6.1.1 Sampling. - Forty (40) detonators shall be subjected to the test specified in 4.4.7.1. Observation shall be made for the functioning time, and all duds thjt may occur will be recorded. Duds will be replaced by detonators from the sample of 200.

6.6.1.2 Rejection. - The lot shall be rejected if the mean functioning time (X) plus 2.03 times the standard deviation (s) exceeds 50 microseconds. Duds shall not be used in the calculation for standard deviation.

6.6.2 Waterproofness, Output and Duds

6.6.2.1 Sampling. - Fifty (50) detonator assemblies shall be subjected to the test specified in 4.4.7.2. Observation shall be made of the waterproofness and output, and duds that may occur will be recorded. Duds will be replaced by detonators from the sample of 200.

6.6.3 Duds

6.6.3.1 Sampling. - The remainder of the 200 detonator assemblies selected in 6.6 shall be subjected to the test specified in 4.4.7.3.

6.6.3.2 Rejection. - The lot shall be rejected if the number of detonator assemblies that failed to fire on the first attempt (duds) accumulated from 6.6.1.1, 6.6.2.1, 6.6.3.1 equals or exceeds four (4).

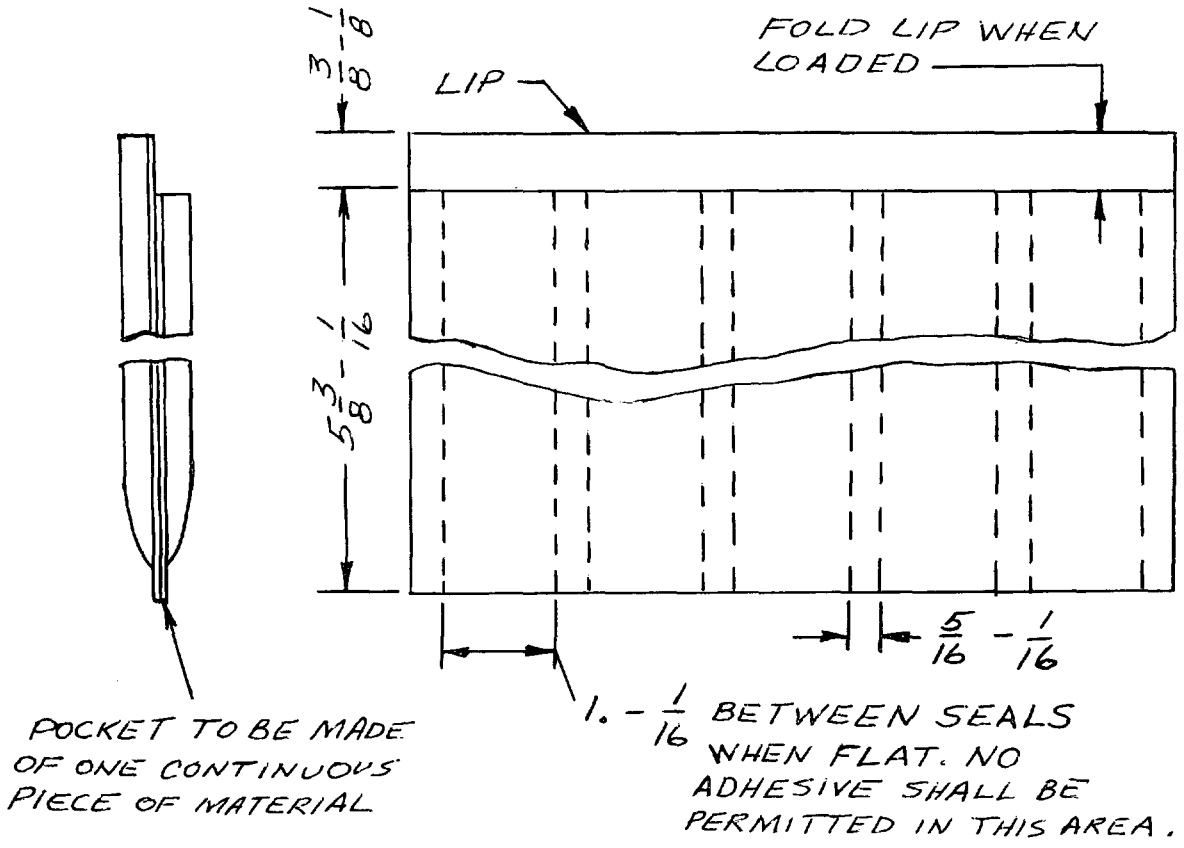
6.7 Supersession Data. - This specification includes the requirements of PA-PD-2917 dated 30 January 1967.

Custodian:
Army-MU

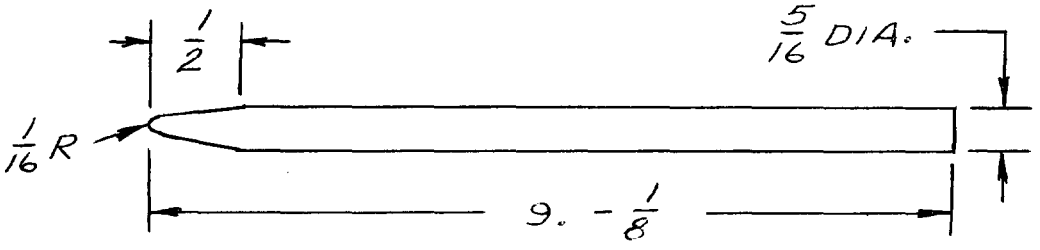
Preparing Activity:
Army-MU

Project Number: 1375-A-026

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POCKET ASSEMBLY



DOWEL SHALL BE SMOOTH WITH NO SHARP CORNERS.

DOWEL

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Department of the Army
Picatinny Arsenal
Dover, New Jersey 07801

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DEPARTMENT OF THE ARMY
DOD 314

OFFICIAL BUSINESS

Commander
Picatinny Arsenal
ATTN: SMUPA-QA-A
Dover, New Jersey 07801

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